

Internet Appendix to

Does Option Trading Have a Pervasive Impact on Underlying Stock Prices?

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This internet appendix contains additional summary statistics and results that supplement the summary statistics and results in the paper.

1. Additional Summary Statistics

Table 1 provides summary statistics for the subsamples of large and not-large and liquid and illiquid firms used in Table 5 of the main text. A liquid stock is defined to be one with an average relative bid-ask spread less than the median, while an illiquid stock is one with an average bid-ask spread greater than the median. Large firms are defined to be the 200 optionable stocks with the greatest average stock market capitalization during the sample period; the not-large firms are all others. The descriptive statistics are first calculated for each underlying stock, and then the averages across the underlying stocks are reported. MM indicates market makers are assumed to delta hedge, MF indicates that both market makers and firm proprietary traders delta hedge, $|r_t|$ is the daily stock absolute return, and $unNormNet\Gamma_t$ and $net\Gamma_t$ are delta hedgers' unnormalized and normalized net position gammas, respectively. $Hedge\Gamma_{t-\tau,t}$ is the component of $net\Gamma_t$ due to past stock price changes, and $Info\Gamma_{t-\tau,t}$ is the component of $net\Gamma_t$ due to changes in options positions.

2. Incomplete Open Interest Data

Our open interest estimates for the 2002–2012 period do not capture trading volume on exchanges other than the CBOE and ISE. This introduces a classic errors in variables problem that would ordinarily result in coefficient estimates being biased toward zero (e.g., if the measurement errors are uncorrelated with the covariates), but can cause the coefficient estimates to be biased away from zero if the measurement errors are correlated with the covariates, depending on the correlation structure. The possibility that our key coefficient estimates are biased toward zero is not a significant concern, as this biases against finding the results we find. It is more important to

address the possibility that measurement errors correlated with the covariates might somehow be driving the results.

To examine if the measurement errors affect our results, we carry out analyses in which we separate the sample stocks into those whose options are heavily traded on the CBOE and ISE and those whose options are not heavily traded on CBOE and ISE. Measurement errors are likely to be smaller for the first group, and larger for the second group. If the measurement errors are driving the results then we expect the coefficients on the key variables to be larger (more negative) for the second group in which the measurement errors are larger; on the other hand, if the measurement errors are simply biasing the estimated coefficients toward zero then we expect the coefficient estimates to be larger (more negative) in the first group in which the measurement errors are smaller.

We measure a stock's option trading on the CBOE and ISE using the ratio of the combined volumes of public customers and firm proprietary traders on the CBOE and ISE to the trading volume reported in OptionMetrics, which captures the option trading volume on all exchanges. For each stock and date, we compute the ratio of the sum of option trading volume on the CBOE and ISE to total option trading volume from OptionMetrics, and then compute the time-series average for each stock. We define a stock to be lightly (heavily) traded on the CBOE and ISE if its average ratio of CBOE and ISE trading volume to total trading volume is lesser (greater) than the cross-sectional median ratio. Panels A and B of Table 2 present the results when we split samples into those lightly and heavily traded on the CBOE and ISE. The results show that the components of net gammas are negatively related to subsequent realized absolute returns for both groups of stocks. The coefficient estimates for stocks heavily traded on the CBOE and ISE are larger (more negative). When market makers are assumed to be the delta hedgers, the coefficient estimates on the key gamma variable in the first column of results are -0.424 (t -statistic -3.41) for stocks with options heavily traded on the CBOE and ISE, and -0.275 (t -statistic -6.41) for stocks with options lightly traded on the CBOE and ISE. We obtain similar results when both market makers and firm proprietary traders are assumed to delta hedge. These results are consistent with measurement error biasing the coefficient estimates toward zero and inconsistent with measurement error driving the results.

The data from the later (2002–2012) sample period do not include information about early exercises of options, creating a different source of measurement error. Note that gammas of ITM

options that are exercised early are zero or close to zero, so in most cases our inability to capture early exercise has no or very limited impact on our estimates of gamma. Our inability to capture early exercises matters only when options become ITM, are exercised, and the stock price subsequently moves so that the exercised options would have been near-the-money had they not been exercised. The main results use gamma estimates that include the gammas of these options that have been exercised, introducing some error into the gamma estimates.

We address this concern by re-estimating the main results after we set open interest of options likely to be exercised early to zero for dates on or after the options were likely to be early exercised. We consider an in-the-money put likely to be exercised early if its mid-price is less than or equal to the exercise value on any trading day, and an in-the-money call likely to be exercised early if its mid-price is less than or equal to the exercise value right before an ex-dividend day. Panel C of Table 2 reports the results. Compared to the results in Table 2 of the main, after adjusting for possible early exercise the coefficient estimate on the variable $net\Gamma_t(t - \tau, S_t) - net\Gamma_t(t - \tau, S_{t-\tau})$ changes from -0.329 to -0.297 when we assume that market makers are the delta hedgers (the second column) and from -0.127 to -0.112 when we assume that firm proprietary traders also delta hedge (the fourth column). The estimated coefficient on the variable $net\Gamma_t - net\Gamma_t(t - \tau, S_t)$ changes from -0.290 to -0.288 in the second column and remains 0.136 in the fourth column. These results indicate that the adjustment for possible early exercise does not have an important impact on the results.

3. Effect of Hedge Rebalancing on Directional Stock Price Movements

To investigate how delta hedge rebalancing affects the directional movement of stock prices, we first compute the normalized net delta of market makers on day t based on their net open interest and stock price as follows:

$$netDelta_t(t, S_t) = 100 \left(\frac{1}{M_t} \right) \times \sum_{j=1}^{N_t} netOpenInterest_{j,t} \Delta_j(t, S_t),$$

where N_t , S_t , and M_t are respectively the number of option series available, the underlying stock price, and the number of shares outstanding as of the close of trading on date t ; the ' t ' in parentheses in $netDelta_t(t, S_t)$ denotes the date on which we measure net open interest.

The quantity of shares market makers need to trade is the negative of the change in delta. We define $dNetDelta$ to be the change in $netDelta$ from $t - \tau$ to t and then decompose $dNetDelta$

into the change due to stock price movement and the change due to option position as we did with the net position gamma:

$$dNetDelta_{t,\tau} = [netDelta_t(t - \tau, S_t) - netDelta_t(t - \tau, S_{t-\tau})] \\ + [netDelta_t(t, S_t) - netDelta_t(t - \tau, S_t)],$$

where $netDelta_t(t - \tau, S_t)$ is market makers' (MMs') net delta based on their net open interest on day $t - \tau$ and the stock price on day t , and $netDelta_t(t - \tau, S_{t-\tau})$ is MMs' net delta based on their net open interest and the stock price on day $t - \tau$. In the above equation, the changes in delta due to stock price movement and option position changes are in the first and the second brackets, respectively. We define a new variable $HedgeDeltaOI_{t-\tau,t}$ as:

$$HedgeDeltaOI_{t-\tau,t} \equiv -[netDelta_t(t - \tau, S_t) - netDelta_t(t - \tau, S_{t-\tau})].$$

This variable measures the option market maker stock order imbalance due to hedge rebalancing arising from past stock price movements. Similarly we define a second new variable $InfoDeltaOI_{t-\tau,t}$ as:

$$InfoDeltaOI_{t-\tau,t} \equiv -[netDelta_t(t, S_t) - netDelta_t(t - \tau, S_t)].$$

This variable measures the stock order imbalance due to changes in market maker positions that need to be delta hedged.

Call and put deltas increase with stock prices. When MM gamma is negative, i.e., when MMs collectively have written options, an increase in the stock price causes their delta to decrease. MMs buy stocks to maintain delta-neutral positions, and stock prices are likely to continue to increase due to this buying pressure. Under this scenario, there is a positive relation between the MM stock order imbalance and the contemporaneous stock return. If MM hedge rebalancing pushes the stock price too high and the stock price reverses afterward, there can be a negative relation between MM stock order imbalance and the subsequent stock return, but this reversal does not necessarily occur quickly. (In fact, Figure 1 shows that the reversals are slow.)

When MMs collectively hold a positive gamma position, MM option delta increases after the stock price increases, and MMs sell stocks to maintain delta neutral positions. Under this scenario, the contemporaneous relation between the MM stock order imbalance and the contemporaneous stock return is not clear, because MM trading will tend to keep the stock price stable, that is to keep the return close to zero. Panel A of Table 3 summarizes this discussion.

We estimate regressions models in which we use dummy variables to indicate if MMs have negative or positive gamma positions. Specifically, we estimate the following regression

specification to examine how the part of option market makers' stock order imbalance due to the stock price change (*HedgeDeltaOI*) and the part due to option position change (*InfoDeltaOI*) relate to contemporaneous stock returns:

$$r_{t-\tau,t} = a + b \times \text{HedgeDeltaOI}_{t-\tau,t} \times I^{G-} + c \times \text{HedgeDeltaOI}_{t-\tau,t} \times I^{G+} \\ + d \times \text{InfoDeltaOI}_{t-\tau,t} \times I^{G-} + e \times \text{InfoDeltaOI}_{t-\tau,t} \times I^{G+} + r_{t-2\tau,t-\tau} + \varepsilon_t,$$

where $r_{t-\tau,t}$ is the stock return from day $t - \tau$ to day t , I^{G-} is one if the average net gamma of MMs is negative from day $t - \tau$ to t and zero otherwise, and I^{G+} is one if the average net gamma of MMs is positive from day $t - \tau$ to t and zero otherwise. We expect the estimate of the coefficient b to be positive. In the empirical tests, we also replace the contemporaneous returns $r_{t-\tau,t}$ with subsequent one or two-day returns to examine if hedge rebalancing leads to return reversals.

The results are reported in Table 3, where τ is five trading days.¹ The coefficients on $\text{HedgeDeltaOI} \times I^{G-}$ are positive and statistically significant when the dependent variable is the contemporaneous return $r_{t-5,t}$ for both the 1990-2001 and 2002-2012 sample periods. This result is consistent with the positive feedback effect between the stock return and the MM stock order imbalance when the MM net position gamma is negative. An increase in the stock price causes the MM delta to decrease, and MM stock purchases to rebalance their hedges drive up the stock price. The coefficient remains positive and statistically significant when the dependent variable is the subsequent one-day return ($r_{t,t+1}$) but with much smaller magnitude (the coefficient estimate is less than 4% as large) during 1990-2001. This suggests that some of the hedge rebalancing occurs on the following day and alters the stock price on that day. For the 2002-2012 sample period, the point estimate in the regression explaining one day ahead returns is positive but small and insignificant. When the dependent variable is the subsequent second day return ($r_{t+1,t+2}$) the coefficient on $\text{HedgeDeltaOI} \times I^{G-}$ becomes negative, -0.022 (t -statistic -1.92) for the first period and -0.016 (t -statistic -1.86) for the second period. The reversed sign is consistent with stock price reversion after MMs hedge rebalancing distorts the stock price.

The coefficient on the interaction between *HedgeDeltaOI* and the positive gamma dummy ($\text{DeltaOI} \times I^{G+}$) is negative for the first period and only marginally significant (-1.201 , t -statistic -1.92), and positive and insignificant for the second period (0.011 , t -statistic 1.49) when the

¹ The results are qualitatively similar if τ is three rather than five trading days.

dependent variable is the contemporaneous return, $r_{t-5,t}$. Thus, there is no convincing evidence that *HedgeDeltaOI* relates to stock returns when gamma is positive. As explained above, this is unsurprising.

The coefficients on MM *InfoDelta* are all negative in the regressions explaining contemporaneous returns and positive in the regressions explaining subsequent returns regardless of the sign of MM gamma. This result suggests that options investors (non-market makers) reduce positive exposure to stocks when stock returns are high, and their trading contains information about subsequent daily stock price movements. Similar results are reported in Hu (2014).

The results in Table 3 are consistent with MM hedge rebalancing altering stock prices. However, because stock returns and option delta are endogenously related, we cannot identify how much of the observed relation (correlation) is driven by the price impact of hedge rebalancing. Our results using estimates of MM gamma in the main text do not suffer from this shortcoming.

4. Effect of Hedging on Unsigned Volumes in Equity Markets

We expect the absolute values of the components of the position gamma to positively relate to unsigned stock volumes. When the position gammas of MMs or other delta hedgers are large in absolute value, the delta hedgers need to trade more stock to remain delta-neutral. Table 4 shows a positive relation between the absolute values of the three components of the position gamma and subsequent stock volumes. The coefficient estimate on the absolute value of the first component of market maker position gamma, $|Hedge\Gamma_{t-\tau,t}|$, is 101.885 (46.333) for the 1990-2001 (2002-2012) sample period. A one standard variation change in $|Hedge\Gamma_{t-\tau,t}|$ is associated with a 0.0714 (0.0832) unit change in log volume, equivalent to 4.8% (5.7%) of the standard deviation of the log stock volume for the period 1990-2001 (2002-2012).²

5. When and Whether Option-implied Volatility Detects Changes in Volatility

Ex-ante, we expect that option market makers will have some but not complete information about the net gammas of market makers and other delta hedgers in aggregate,³ and adjust option prices based on the information they have. Table 5 shows that implied volatility on date t in the

² The standard deviations of $|Hedge\Gamma_{t-5,t}|$ are 0.0070 and 0.0018 for the first and second periods, respectively, and $0.0714 = 101.885 \times 0.0007$ and $0.0832 = 46.333 \times 0.0018$. The standard deviations of log stock volume are 1.4676 and 1.4679 for the first and second periods, respectively, $0.0174/1.4676 = 4.8\%$, and $0.0832/1.4679 = 5.7\%$.

³ Each options market maker knows its own positions and not the positions of other market makers and delta hedgers. But options market makers may be able to use information about options order flow to develop estimates of the positions of other market makers.

first period and from dates $t - 2$ to t in the second period helps predict the absolute return on date $t + 1$. After controlling for implied volatility, the components of the net position gamma still have predictive power but with the somewhat smaller magnitudes of the coefficient estimates. For example, the coefficient estimates on the first component of MM position gamma (that is, hedge gamma) change from -0.576 to -0.508 for the first period, and from -0.325 to -0.262 for the second period. These results are consistent with the changes in volatility being partially, but not completely, reflected in implied volatilities, which is what we expect.

6. More on Return Reversals

A large absolute return is more likely to reverse than a small one. In another test for return reversals, we focus on stocks with large absolute returns and examine whether return reversals are related to $Hedge\Gamma_{t-5,t}$ or $Info\Gamma_{t-5,t}$. For each date t , we take stocks in the highest and lowest return deciles on date $t + 1$, and then sort them into deciles based on $Hedge\Gamma_{t-5,t}$ or $Info\Gamma_{t-5,t}$. We then compute the date $t + 1$ to $t + 15$ cumulative returns of the stocks in the lowest and highest gamma deciles within the high and low return deciles and plot the returns in Figure 1(a) for $Hedge\Gamma_{t-5,t}$ and Figure 1(b) for $Info\Gamma_{t-5,t}$. Figure 1(a) shows that low $Hedge\Gamma_{t-5,t}$ portfolios have more extreme returns on day $t + 1$ than high hedge gamma portfolios. (The date $t + 1$ returns of low hedge gamma stocks are lower than those of high hedge gamma stocks in the two left-hand panels, and larger in the right-hand panels.) This is consistent with the hypothesis that hedge rebalancing of negative gamma positions causes returns to be more extreme, that is it causes negative (positive) returns to be more negative (positive). One can also see that the returns of the low hedge gamma portfolios on later days become less extreme than (or the same as) those of high hedge gamma portfolios, indicating that stocks with lower $Hedge\Gamma_{t-5,t}$ have greater return reversals.

When portfolios are sorted on $Info\Gamma_{t-5,t}$, as shown in Figure 1(b), the returns of low $Info\Gamma_{t-5,t}$ stocks are also more extreme than those of high information gamma, which can occur either because of hedge rebalancing or because investors tend to buy rather than sell options when they expect stock prices to change a lot. Figure 1(b) also shows a certain degree of return reversal especially when stocks have extreme negative returns on day $t + 1$, but shows no difference in the magnitudes of return reversal between high and low $Info\Gamma_{t-5,t}$ portfolios. Overall, Figure 1(a) and

Figure 1(b) provide evidence with the hypothesis that stocks have temporarily price movements arising from delta hedging rebalancing of options market makers.

7. Possible alternative explanation: A non-linear relation between past returns and future absolute returns.

One might hypothesize that the correlation between net position gamma and future absolute returns is driven by a non-linear relation between past and future absolute returns. The main regression specification in the paper (equation 8) includes interactions between the absolute returns and the dummy variables for positive returns to allow for different impacts of positive and negative returns on volatility, reducing the likelihood that this is the case. We carry out two additional analyses to further address this possible concern.

First, we use a control sample matched on $r_{t-\tau,t}$. For each trading day, we sort stocks into terciles based on the absolute value of hedge gamma, that is $|Hedge\Gamma_{t-\tau,t}|$. For each stock in the high absolute hedge gamma tercile, we locate a control stock in the lowest absolute hedge gamma tercile that has return closest to the return of the high tercile stock over the period from day -5 to day 0 . We then compute the difference between the absolute returns $|r_{t+1}|$ of the high and low tercile stocks, and use this as the dependent variable in regressions estimated using the stocks in the high hedge gamma tercile. Table 6 in this internet appendix reports results based on gamma computed assuming that market makers are the delta hedgers. It shows that after controlling for the directional effect of $r_{t-\tau,t}$ using the matched sample, the coefficients on the components of position gamma remain negative. We obtain similar results when we include firm proprietary traders as delta hedgers.

Second, we let the net public option open interest be one for all stocks on all days and re-estimate the main regression specification. If the results are driven by a relation between $r_{t-\tau,t}$ and $|r_{t+1}|$, then the coefficients on components of position gamma will be negative. As shown in Table 7 of this appendix, this is not the case. The coefficient estimates on $Hedge\Gamma_{t-\tau,t}$ are not statistically different from zero, suggesting that our results are not driven by a relation between $r_{t-\tau,t}$ and $|r_{t+1}|$.

8. Unnormalized Net Option Position Gamma

The regressions in the main text that estimate the relation between the net position gamma and subsequent absolute returns use net position gammas normalized by S/M , where S is the

underlying stock price and M is the number of shares outstanding. As discussed in the main text, this normalization is either required or suggested by the theoretical literature cited in the main text. It also makes the normalized gamma dimensionless. Table 8 verifies that the main results are not driven by this normalization by presenting regression results in which unnormalized net position gammas are used as the key covariates. The coefficients on the components of gamma are negative and highly significant, consistent with the results based on normalized gammas.

9. Alternative Specification in which the Dependent Variable is the First Difference of the Absolute Return

To examine whether the main results are possibly driven by the persistence of $|r_t|$, we estimate regressions in which the dependent variable is the first difference in the absolute return $\Delta|r_{t+1}| = |r_{t+1}| - |r_t|$ and the lagged returns included as control variables are also replaced by their first differences $\Delta|r_{t-i}| = |r_{t-i}| - |r_{t-i-1}|$, for $i = 0, \dots, 9$. The results are reported in Table 9. One can see that the coefficients on the first and third components of gamma (hedge and information gamma) are still negative and significant in both periods. For example, when we consider the options market makers to be the delta hedgers, the coefficient estimate on $Hedge\Gamma_{t-\tau,t}$ is -0.369 (t -statistic -4.26) for the first period, and -0.214 (t -statistic -3.68) for the second period. For the second component of gamma, i.e., the position gamma from five days in the past, the coefficient estimate becomes positive and significant, though the magnitude of the coefficient estimate is small.

10. Cross-sectional Average Gamma and S&P500 Index Absolute Returns

The summary statistics in Table 1 of the main text indicate that market maker net gamma was on average positive over the 1990–2001 sample period and negative but close to zero over the 2002–2012 period. If there is correlation across stocks in the market makers' net gammas so that gammas are generally negative or positive at certain times, hedge rebalancing might increase (reduce) the volatility of the aggregate stock market. To test this hypothesis, we use the cross-sectional average gammas of the largest 500 optionable stocks to examine whether they help explain S&P 500 index absolute daily returns.

Table 10 of this internet appendix contains the summary statistics of the variables used in these regression analyses. The results are reported in Table 11. The coefficient estimates on the cross-sectional average of the first component of gamma, $avgHedge\Gamma_{t-\tau,t}$, are negative and

significant, with the estimates being -1.017 (t -statistic -2.69) and -0.880 (t -statistic -2.11) for the first and second periods, respectively. These results are consistent with the hypothesis that options market maker delta hedging rebalancing also affects the subsequent volatility of the S&P 500 index. The economic magnitude of the effect is smaller than that for the individual stocks, which is unsurprising. The average daily absolute return of the S&P 500 index in the first (second) sample period is 70 (89) basis points, and the standard deviation of the average of the first component of gamma is 0.0004 (0.0013) for the first (second) period. Thus, a one standard deviation shock to the average gamma variable related to hedging rebalancing has a 4.6% ($= 0.808 \times 0.0004 / 0.007$) impact on the daily absolute return of the S&P 500 index in the first sample period and a 7.0% ($= 0.476 \times 0.0013 / 0.0089$) impact in the second period. For individual stocks, the corresponding impacts based on the results in Table 2 of the main text are 10.5% and 13.3% of the daily absolute returns of optionable stocks in the two sample periods.

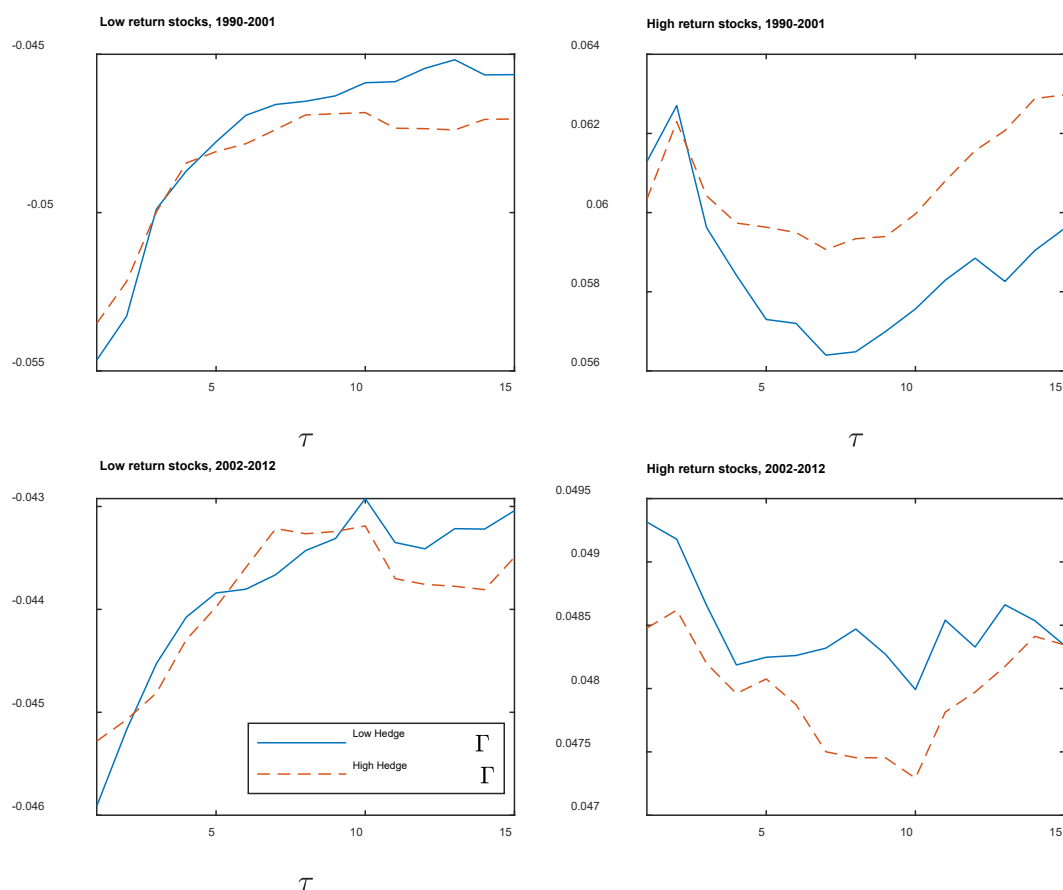


Figure 1(a)

Cumulative returns for stocks in low and high return/hedge gamma deciles

Stocks are first sorted into deciles based on their day $t + 1$ returns, then stocks in the lowest and the highest return deciles are further sorted into deciles using market maker Hedge Γ . The figure displays the cumulative stock returns from day $t + 1$ to $t + \tau$, where $\tau = 1, 2, \dots, 15$, for stocks in the lowest and/or highest return and Hedge Γ deciles.

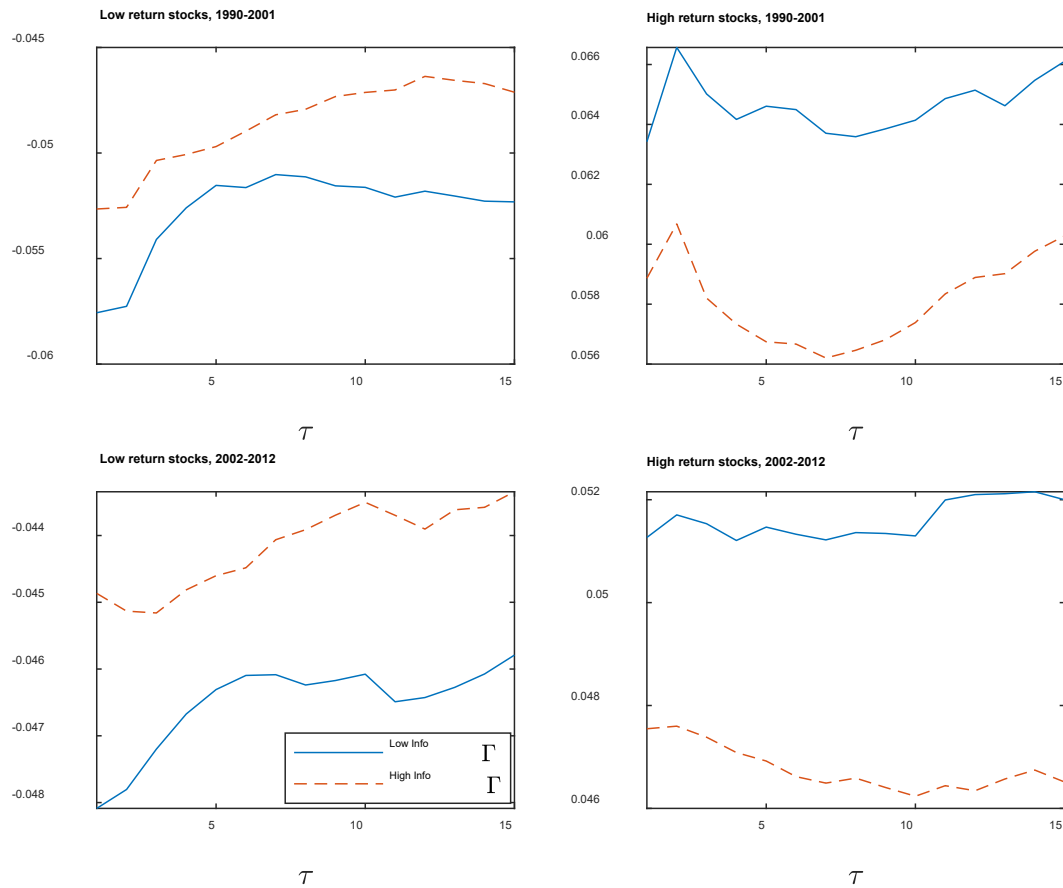


Figure 1(b)

Cumulative returns for stocks in low and high return/information gamma deciles

Stocks are first sorted into deciles based on returns on day $t + 1$, then stocks in the lowest and the highest return deciles are further sorted into deciles on market maker $Info\Gamma$. The figure displays the cumulative stock returns from day $t + 1$ to $t + \tau$, where $\tau = 1, 2, \dots, 15$, for stocks in the lowest and/or highest return and $Info\Gamma$ deciles.

Table 1**Descriptive statistics for stocks of different size and liquidity**

Means, standard deviations, extrema, and quantiles for unnormalized and normalized gamma and the hedge and information gamma variables for the subsamples of large and not-large and liquid and illiquid firms. A liquid stock is defined to be one with an average relative bid-ask spread less than the median, while an illiquid stock is one with an average bid-ask spread greater than the median. Large firms are defined to be the 200 optionable stocks with the greatest average stock market capitalization during the sample period; the not-large firms are all others. The descriptive statistics are first calculated for each underlying stock, and then the averages across the underlying stocks are reported. MM indicates that market makers are the delta hedgers, MF indicates market makers and firm proprietary traders are delta hedgers, $|r_t|$ is the daily stock absolute return, and $unNormNet\Gamma_t$ and $net\Gamma_t$ are delta hedgers' unnormalized and normalized net position gammas, respectively. $Hedge\Gamma_{t-\tau,t}$ is the component of $net\Gamma_t$ due to past stock price changes, and $Info\Gamma_{t-\tau,t}$ is the component of $net\Gamma_t$ due to change in options positions.

A. Largest 200 stocks									
1990–2001	Mean	Std.	Min.	10%	25%	50%	75%	90%	Max.
$ r_t $	0.0226	0.0241	0.0000	0.0031	0.0080	0.0172	0.0308	0.0479	0.2091
$unNormNet\Gamma_t$ MM	51,772	64,552	–721,056	–3,385	15,927	39,440	74,241	127,940	356,108
$unNormNet\Gamma_t$ MF	68,821	89,403	–821,378	–6,259	20,088	50,359	97,066	171,721	507,879
$Net\Gamma_t$ MM	0.0033	0.0043	–0.1503	–0.0007	0.0010	0.0027	0.0051	0.0080	0.0295
$Net\Gamma_t$ MF	0.0042	0.0055	–0.1705	–0.0009	0.0013	0.0035	0.0064	0.0102	0.0375
$Hedge\Gamma_{t-\tau,t}$ MM	0.0000	0.0022	–0.1089	–0.0015	–0.0004	0.0001	0.0005	0.0015	0.0223
$Hedge\Gamma_{t-\tau,t}$ MF	0.0000	0.0027	–0.1313	–0.0018	–0.0005	0.0001	0.0006	0.0018	0.0282
$Info\Gamma_{t-\tau,t}$ MM	0.0000	0.0032	–0.1718	–0.0022	–0.0008	0.0002	0.0010	0.0022	0.0256
$Info\Gamma_{t-\tau,t}$ MF	0.0000	0.0032	–0.2078	–0.0030	–0.0009	0.0002	0.0012	0.0023	0.0332
2002–2012									
$ r_t $	0.0149	0.0170	0.0000	0.0018	0.0046	0.0102	0.0193	0.0324	0.2122
$nonNormNet\Gamma_t$ MM	–69,477	188,691	–10,307,612	–325,108	–140,340	–36,821	38,342	106,513	453,739
$nonNormNet\Gamma_t$ MF	72,976	254,238	–9,570,022	–207,327	–56,035	47,605	187,443	374,228	1,160,888
$Net\Gamma_t$ MM	–0.0067	0.0071	–0.2820	–0.0053	–0.0012	0.0005	–0.0009	0.0021	0.0189
$Net\Gamma_t$ MF	–0.0025	0.0109	–0.3684	–0.0064	–0.0053	–0.0023	0.0032	0.0085	0.0474
$Hedge\Gamma_{t-\tau,t}$ MM	0.0001	0.0047	–0.0624	–0.0022	–0.0010	0.0000	0.0012	0.0026	0.0434
$Hedge\Gamma_{t-\tau,t}$ MF	0.0000	0.0057	–0.0656	–0.0032	–0.0011	0.0000	0.0013	0.0028	0.0619
$Info\Gamma_{t-\tau,t}$ MM	–0.0001	0.0061	–0.0481	–0.0025	–0.0015	–0.0003	0.0011	0.0032	0.0591
$Info\Gamma_{t-\tau,t}$ MF	0.0000	0.0077	–0.0531	–0.0032	–0.0013	0.0000	0.0017	0.0042	0.0724

B. Not-large stocks

1990–2001	Mean	Std.	Min.	10%	25%	50%	75%	90%	Max.
$ r_t $	0.0307	0.0322	0.0000	0.0033	0.0098	0.0229	0.0423	0.0663	0.3135
$unNormNet\Gamma_t$ MM	9,623	18,305	-416,586	-6,645	60	6,381	16,172	30,923	98,660
$unNormNet\Gamma_t$ MF	10,711	22,314	-514,815	-8,294	-140	6,893	17,836	35,299	124,646
$Net\Gamma_t$ MM	0.0034	0.0073	-0.2094	-0.0032	-0.0003	0.0024	0.0062	0.0116	0.0408
$Net\Gamma_t$ MF	0.0037	0.0086	-0.2579	-0.0039	-0.0004	0.0025	0.0067	0.0128	0.0487
$Hedge\Gamma_{t-\tau,t}$ MM	0.0000	0.0057	-0.2189	-0.0021	-0.0005	0.0000	0.0006	0.0021	0.0342
$Hedge\Gamma_{t-\tau,t}$ MF	0.0000	0.0053	-0.3431	-0.0023	-0.0006	0.0000	0.0007	0.0031	0.0405
$Info\Gamma_{t-\tau,t}$ MM	0.0000	0.0051	-0.2718	-0.0032	-0.0007	0.0002	0.0012	0.0031	0.0351
$Info\Gamma_{t-\tau,t}$ MF	0.0000	0.0059	-0.2848	-0.0035	-0.0008	0.0002	0.0013	0.0033	0.0439
2002–2012									
$ r_t $	0.0222	0.0251	0.0000	0.0025	0.0067	0.0152	0.0290	0.0485	0.3145
$unNormNet\Gamma_t$ MM	-4,479	37,250	-2,307,612	-39,759	-16,579	-2,247	9,689	27,335	145,493
$unNormNet\Gamma_t$ MF	3,758	50,415	-2,570,022	-38,662	-13,391	1,337	17,956	48,723	260,094
$Net\Gamma_t$ MM	-0.0013	0.0134	-0.5167	-0.0073	-0.0014	0.0002	0.0040	0.0111	0.0517
$Net\Gamma_t$ MF	-0.0022	0.0177	-0.7819	-0.0102	-0.0069	-0.0009	0.0044	0.0123	0.0650
$Hedge\Gamma_{t-\tau,t}$ MM	0.0000	0.0105	-0.1181	-0.0034	-0.0008	0.0000	0.0009	0.0035	0.0864
$Hedge\Gamma_{t-\tau,t}$ MF	0.0000	0.0129	-0.1432	-0.0042	-0.0010	0.0000	0.0011	0.0041	0.1193
$Info\Gamma_{t-\tau,t}$ MM	-0.0002	0.0096	-0.0927	-0.0043	-0.0013	0.0000	0.0010	0.0041	0.1077
$Info\Gamma_{t-\tau,t}$ MF	-0.0001	0.0150	-0.1389	-0.0058	-0.0014	0.0000	0.0014	0.0052	0.1394

C. Liquid stocks

1990–2001	Mean	Std.	Min.	10%	25%	50%	75%	90%	Max.
$ r_t $	0.0278	0.0278	0.0000	0.0023	0.0082	0.0180	0.0350	0.0000	0.2729
$unNormNet\Gamma_t$ MM	34,972	34,972	-863,202	-8,996	3,225	17,887	30,437	55,086	168,426
$unNormNet\Gamma_t$ MF	46,827	46,829	-981,489	-10,320	4,205	20,501	38,125	68,877	237,003
$Net\Gamma_t$ MM	0.0049	0.0053	-0.1503	-0.0019	0.0004	0.0024	0.0054	0.0094	0.0323
$Net\Gamma_t$ MF	0.0059	0.0059	-0.1805	-0.0025	0.0004	0.0028	0.0063	0.0103	0.0388
$Hedge\Gamma_{t-\tau,t}$ MM	0.0029	0.0032	-0.1289	-0.0018	-0.0005	0.0001	0.0006	0.0015	0.0294
$Hedge\Gamma_{t-\tau,t}$ MF	0.0036	0.0037	-0.1913	-0.0021	-0.0006	0.0001	0.0007	0.0021	0.0336
$Info\Gamma_{t-\tau,t}$ MM	0.0043	0.0026	-0.1718	-0.0028	-0.0008	0.0002	0.0011	0.0026	0.0319
$Info\Gamma_{t-\tau,t}$ MF	0.0053	0.0039	-0.2078	-0.0033	-0.0008	0.0002	0.0012	0.0022	0.0353
2002–2012									
$ r_t $	0.0205	0.0212	0.0000	0.0021	0.0055	0.0124	0.0236	0.0395	0.2980
$unNormNet\Gamma_t$ MM	63,894	63,894	-4,073,048	-86,872	-36,894	-7,202	14,126	39,407	243,372
$unNormNet\Gamma_t$ MF	85,490	85,490	-4,872,215	-68,195	-21,233	7,540	42,024	97,658	503,975
$Net\Gamma_t$ MM	0.0090	0.0080	-0.2433	-0.0038	-0.0063	-0.0016	0.0015	0.0070	0.0357
$Net\Gamma_t$ MF	0.0129	0.0083	-0.5501	-0.0063	-0.0054	-0.0003	0.0039	0.0080	0.0462
$Hedge\Gamma_{t-\tau,t}$ MM	0.0059	0.0052	-0.0792	-0.0024	-0.0007	0.0000	0.0008	0.0027	0.0619
$Hedge\Gamma_{t-\tau,t}$ MF	0.0073	0.0073	-0.0844	-0.0032	-0.0008	0.0000	0.0009	0.0033	0.0789
$Info\Gamma_{t-\tau,t}$ MM	0.0065	0.0065	-0.0652	-0.0039	-0.0012	0.0000	0.0009	0.0031	0.0870
$Info\Gamma_{t-\tau,t}$ MF	0.0084	0.0082	-0.0831	-0.0046	-0.0013	0.0001	0.0013	0.0041	0.1036

D. Illiquid stocks									
1990–2001	Mean	Std.	Min.	10%	25%	50%	75%	90%	Max.
$ r_t $	0.0338	0.0362	0.0000	0.0030	0.0126	0.0251	0.0470	0.0707	0.3551
$unNormNet\Gamma_t$ MM	3,458	9,846	−183,622	−3,780	−99	1,467	12,803	23,844	72,740
$unNormNet\Gamma_t$ MF	2,708	10,497	−381,645	−7,377	−779	1,845	13,363	27,301	83,527
$Net\Gamma_t$ MM	0.0036	0.0091	−0.2503	−0.0039	−0.0006	0.0024	0.0068	0.0128	0.0417
$Net\Gamma_t$ MF	0.0037	0.0101	−0.3135	−0.0045	−0.0007	0.0025	0.0080	0.0153	0.0512
$Hedge\Gamma_{t-\tau,t}$ MM	0.0001	0.0068	−0.2891	−0.0023	−0.0006	0.0000	0.0007	0.0043	0.0386
$Hedge\Gamma_{t-\tau,t}$ MF	0.0001	0.0063	−0.4227	−0.0024	−0.0006	0.0000	0.0007	0.0044	0.0474
$Info\Gamma_{t-\tau,t}$ MM	0.0000	0.0074	0.2379	−0.0034	−0.0007	0.0002	0.0013	0.0043	0.0421
$Info\Gamma_{t-\tau,t}$ MF	−0.0001	0.0081	−0.3122	−0.0035	−0.0008	0.0002	0.0013	0.0047	0.0607
2002–2012									
$ r_t $	0.0255	0.0288	0.0000	0.0028	0.0076	0.0175	0.0334	0.0556	0.3285
$unNormNet\Gamma_t$ MM	−2,528	35,770	−1,654,892	−27,198	−9,925	−1,112	8,418	24,051	115,322
$unNormNet\Gamma_t$ MF	997	47,112	−1,153,793	−27,997	−10,251	246	12,616	35,938	187,381
$Net\Gamma_t$ MM	0.0015	0.0166	−0.7507	−0.0080	−0.0108	−0.0025	0.0039	0.0129	0.0663
$Net\Gamma_t$ MF	−0.0033	0.0197	−0.9819	−0.0143	−0.0105	−0.0017	0.0046	0.0164	0.0778
$Hedge\Gamma_{t-\tau,t}$ MM	0.0000	0.0128	−0.1568	−0.0043	−0.0010	0.0000	0.0011	0.0042	0.1101
$Hedge\Gamma_{t-\tau,t}$ MF	0.0000	0.0127	−0.1956	−0.0051	−0.0012	0.0000	0.0012	0.0049	0.1451
$Info\Gamma_{t-\tau,t}$ MM	−0.0002	0.0115	−0.1148	−0.0060	−0.0015	0.0000	0.0012	0.0051	0.1250
$Info\Gamma_{t-\tau,t}$ MF	−0.0002	0.0196	−0.1589	−0.0069	−0.0017	0.0000	0.0015	0.0062	0.1760

Table 2**Regressions of absolute return on components of net gamma to assess whether results are affected by measurement error**

Panels A and B report results from estimating model (8) expressing the absolute return $|r_{t+1}|$ in terms of the components of the normalized net position gamma and lagged returns for stocks with options heavily and lightly traded on the CBOE and ISE, respectively. For each stock and date, we compute the ratio of the sum of option trading volume on the CBOE and ISE to total option trading volume from OptionMetrics. Stocks for which the average ratio is above the median average ratio are defined to be stocks with options heavily traded on the CBOE and ISE, while stocks for which the average ratio is below the median are stocks with options lightly traded on the CBOE and ISE. Panel C reports results for the full sample when we compute the net position gammas after setting the open interest of options likely to be exercised early to zero. An in-the-money put is likely to be exercised early if its mid-price is less than or equal to the exercise value on any trading day, and an in-the-money call is likely to be exercised early if its mid-price is less than or equal to the exercise value right before an ex-dividend day. Standard errors are constructed from a covariance matrix for the average coefficients, which is formed by clustering observations by date and firm with common shocks, a method detailed in Thompson (2011). The associated t -statistics are reported in parentheses. The coefficient estimates on variables related to current and lagged absolute returns are omitted. The sample period is 2002–2012 for all panels.

Market Maker			Market Maker + Firm	
<i>A. (CBOE + ISE volume)/(total volume) below median ratio</i>				
Constant	0.008	(25.99)	0.008	(25.85)
<i>Hedge</i> $\Gamma_{t-\tau,t}$	-0.275	(-6.41)	-0.149	(-5.46)
<i>Net</i> $\Gamma_{t-\tau}$	-0.119	(-7.42)	-0.062	(-7.82)
<i>Info</i> $\Gamma_{t-\tau,t}$	-0.276	(-6.68)	-0.104	(-8.37)
...	...	...	...	...
<i>B. (CBOE + ISE volume)/(total volume) above median ratio</i>				
Constant	0.009	(27.76)	0.009	(27.87)
<i>Hedge</i> $\Gamma_{t-\tau,t}$	-0.424	(-3.41)	-0.209	(-3.72)
<i>Net</i> $\Gamma_{t-\tau}$	-0.251	(-4.51)	-0.096	(-3.65)
<i>Info</i> $\Gamma_{t-\tau,t}$	-0.337	(-6.62)	-0.193	(-5.52)
...	...	...	...	...
<i>C. Adjust positions for possible early exercise</i>				
Constant	0.009	(27.60)	0.009	(27.84)
<i>Hedge</i> $\Gamma_{t-\tau,t}$	-0.297	(-6.65)	-0.112	(-5.76)
<i>net</i> $\Gamma_{t-\tau}$	-0.147	(-4.89)	-0.082	(-5.47)
<i>Info</i> $\Gamma_{t-\tau,t}$	-0.288	(-8.40)	-0.136	(-8.93)
...	...	...	...	...

Table 3**Relation between hedge delta and stock returns**

Panel A summarizes how the change in options market maker (MM) delta due to past stock price movement relates to stock returns. The first two columns of results in Panel B report the results of regressions:

$$r_{t-5,t} = a + b \times \text{HedgeDeltaOI}_{t-5,t} \times I^{G-} + c \times \text{HedgeDeltaOI}_{t-5,t} \times I^{G+} + d \times \text{InfoDeltaOI}_{t-5,t} \times I^{G-} + e \times \text{InfoDeltaOI}_{t-5,t} \times I^{G+} + r_{t-10,t-5} + \varepsilon_t,$$

where $\text{HedgeDeltaOI}_{t-5,t}$ is MM stock order imbalance due to past stock price movement from trade day $t-5$ to t . The indicator variable I^{G-} equals one if the daily average of MM position gamma from day $t-5$ to t is negative and zero otherwise. I^{G+} is the corresponding indicator variable for positive MM position gamma. $\text{InfoDeltaOI}_{t-5,t}$ is option market maker stock order imbalance due to changes in option open interest from trade day $t-5$ to day t . Panel B also reports the results where the dependent variable is the subsequent day ($r_{t,t+1}$) or next day ($r_{t+1,t+2}$) stock return. Standard errors are constructed from a covariance matrix for the average coefficients, which is formed by clustering observations by date and firm with common shocks, a method detailed in Thompson (2011). The associated t -statistics are reported in parentheses next to the coefficient estimates.

A. Predicted relation between stock returns and MM delta

MM gamma < 0	(1) S increases, MM option delta decreases, (2) MMs buy stock
	(3) S decreases, MM option delta increases, (4) MMs sell stock
	(1) $\rightarrow$ (2) $\rightarrow$ (1) $\rightarrow$ (2) ... or (3) $\rightarrow$ (4) $\rightarrow$ (3) $\rightarrow$ (4) ...
	Positive relation between stock return and MM stock order imbalance
MM gamma > 0	(1) S increases, MM option delta increases, (2) MMs sell stock
	(3) S decreases, MM option delta decreases (4) MMs buy stock
	(1) $\rightarrow$ (2) $\rightarrow$ (3) $\rightarrow$ (4) $\rightarrow$... or (3) $\rightarrow$ (4) $\rightarrow$ (1) $\rightarrow$ (2), or (1) $\rightarrow$ (2) $\rightarrow$ (1) $\rightarrow$ (2) ... or (3) $\rightarrow$ (4) $\rightarrow$ (3) $\rightarrow$ (4)
	Relation between S and MM stock order imbalance weak and unclear

B. Empirical results for the relation between stock returns and MM delta

Dependent variable	$r_{t-5,t}$		$r_{t,t+1}$		$r_{t+1,t+2}$	
1990–2001						
Constant	0.003	(2.94)	0.001	(1.72)	0.001	(1.65)
$HedgeDeltaOI \times I^{G-}$	0.791	(8.30)	0.033	(3.12)	−0.022	(−1.92)
$HedgeDeltaOI \times I^{G+}$	−1.201	(−2.71)	−0.037	(−1.23)	0.008	(0.39)
$InfoDeltaOI \times I^{G-}$	−0.093	(−5.00)	0.011	(2.64)	0.002	(0.51)
$InfoDeltaOI \times I^{G+}$	−0.149	(−4.92)	0.007	(1.27)	0.003	(0.40)
$r_{t-10,t-5}$	−0.016	(−0.76)				
$r_{t-1,t}$			−0.017	(−0.77)		
$r_{t,t+1}$					−0.016	(−0.74)
2002–2012						
Constant	0.002	(2.21)	0.000	(1.18)	0.000	(1.12)
$HedgeDeltaOI \times I^{G-}$	0.334	(11.30)	0.013	(0.98)	−0.016	(−1.86)
$HedgeDeltaOI \times I^{G+}$	0.011	(1.49)	0.001	(0.16)	0.006	(0.72)
$InfoDeltaOI \times I^{G-}$	−0.181	(−6.66)	0.024	(3.46)	0.003	(0.42)
$InfoDeltaOI \times I^{G+}$	−0.289	(−8.77)	0.023	(3.71)	0.006	(0.85)
$r_{t-10,t-5}$	−0.027	(−1.23)			−0.029	(−1.29)
$r_{t-1,t}$			−0.025	(−1.13)		
$r_{t,t+1}$					−0.029	(−1.29)

Table 4**Effect of gamma on stock market volume**

The table presents the results of the regressions estimating how the absolute values of the components of net position gamma relate to subsequent (unsigned) stock trading volume. The dependent variable SV_{t+1} is the log of stock market trading volume on day $t + 1$. The variable $|Hedge\Gamma_{t-\tau,t}|$ is the absolute value of the change in delta hedgers' net gamma due to past stock price movement from trade day $t - \tau$ to t , $Net\Gamma_{t-\tau}$ is net position gamma on day $t - \tau$, and $|Info\Gamma_{t-\tau,t}|$ is the change in delta hedgers' net gamma due to changes in open interest from trade day $t - \tau$ to day t . Standard errors are constructed from a covariance matrix for the average coefficients, which is formed by clustering observations by date and firm with common shocks, a method detailed in Thompson (2011). The associated t -statistics are reported in parentheses next to the coefficient estimates.

1990–2001	Market Maker		Market Maker + Firm	
Constant	2.249	(27.28)	2.253	(27.25)
$ Hedge\Gamma_{t-\tau,t} $	101.885	(10.42)	89.481	(9.88)
$ Net\Gamma_{t-\tau} $	16.220	(1.53)	15.501	(1.45)
$ Info\Gamma_{t-\tau,t} $	24.943	(1.91)	14.579	(1.79)
SV_t	0.406	(43.25)	0.406	(43.27)
SV_{t-1}	0.120	(23.15)	0.120	(23.16)
SV_{t-2}	0.077	(20.16)	0.077	(20.16)
SV_{t-3}	0.065	(18.18)	0.065	(18.18)
SV_{t-4}	0.058	(18.30)	0.058	(18.30)
SV_{t-5}	0.042	(16.16)	0.042	(16.15)
SV_{t-6}	0.035	(15.13)	0.035	(15.13)
SV_{t-7}	0.041	(15.23)	0.041	(15.23)
SV_{t-8}	0.047	(16.34)	0.047	(16.34)
SV_{t-9}	0.041	(15.70)	0.041	(15.69)
2002–2012				
Constant	1.886	(26.01)	1.886	(26.00)
$ Hedge\Gamma_{t-\tau,t} $	46.333	(4.36)	34.966	(5.69)
$ Net\Gamma_{t-\tau} $	16.719	(3.68)	6.581	(2.15)
$ Info\Gamma_{t-\tau,t} $	42.058	(5.58)	30.648	(4.87)
SV_t	0.437	(39.39)	0.437	(39.39)
SV_{t-1}	0.122	(19.49)	0.122	(19.49)
SV_{t-2}	0.077	(15.97)	0.077	(15.97)
SV_{t-3}	0.060	(14.54)	0.060	(14.54)
SV_{t-4}	0.054	(14.08)	0.054	(14.08)
SV_{t-5}	0.037	(13.22)	0.037	(13.22)
SV_{t-6}	0.030	(12.54)	0.030	(12.54)
SV_{t-7}	0.038	(13.49)	0.038	(13.49)
SV_{t-8}	0.042	(14.03)	0.042	(14.03)
SV_{t-9}	0.040	(13.97)	0.040	(13.97)

Table 5**Regressions using implied volatility to help explain the absolute return**

The dependent variable is the absolute return on day $t + 1$, $|r_{t+1}|$. $Hedge\Gamma_{t-\tau,t}$ is the change in delta hedgers' net position gamma due to stock price movement from day $t - \tau$ to t , $Net\Gamma_{t-\tau}$ is the net position gamma on day $t - \tau$, and $Info\Gamma_{t-\tau,t}$ is the change in delta hedgers' net position gamma due to changes in open interest. IV is the average implied volatility of ATM options with maturity between 5 trading days and 3 months. Standard errors are constructed from the covariance matrix for the average coefficients, which is formed by clustering observations by date and firm with common shocks (Thompson (2011)). The t -statistics are reported in parentheses next to the coefficient estimates. Coefficient estimates on constant and variables related lagged absolute return from day $t - 9$ to $t - 4$ are omitted to save space.

1990–2001	Market Maker		Market Maker + Firm	
$Hedge\Gamma_{t-\tau,t}$	-0.508	(-6.27)	-0.427	(-6.41)
$Net\Gamma_{t-\tau}$	-0.435	(-5.16)	-0.360	(-5.25)
$Info\Gamma_{t-\tau,t}$	-0.466	(-8.33)	-0.382	(-7.36)
IV_t	0.023	(12.85)	0.023	(12.16)
IV_{t-1}	0.002	(1.15)	0.002	(1.13)
IV_{t-2}	0.002	(1.13)	0.002	(1.33)
IV_{t-3}	-0.001	(-0.47)	-0.001	(-0.82)
IV_{t-4}	0.000	(-0.22)	-0.001	(-0.37)
$ r_t $	0.115	(13.43)	0.115	(13.46)
$ r_t \times I^{r_t>0}$	-0.034	(-3.84)	-0.034	(-3.83)
$ r_{t-1} $	0.064	(7.48)	0.064	(7.52)
$ r_{t-1} \times I^{r_{t-1}>0}$	-0.055	(-6.15)	-0.055	(-6.15)
$ r_{t-2} $	0.039	(5.52)	0.040	(5.56)
$ r_{t-2} \times I^{r_{t-2}>0}$	-0.033	(-4.21)	-0.033	(-4.23)
$ r_{t-3} $	0.036	(5.43)	0.037	(5.47)
$ r_{t-3} \times I^{r_{t-3}>0}$	-0.032	(-4.36)	-0.032	(-4.38)
2002–2012				
$Hedge\Gamma_{t-\tau,t}$	-0.262	(-5.30)	-0.093	(-5.09)
$net\Gamma_{t-\tau}$	-0.109	(-2.57)	-0.061	(-4.06)
$Info\Gamma_{t-\tau,t}$	-0.196	(-5.40)	-0.077	(-5.58)
IV_t	0.027	(10.32)	0.027	(10.50)
IV_{t-1}	0.008	(2.65)	0.007	(2.65)
IV_{t-2}	0.005	(2.10)	0.005	(1.96)
IV_{t-3}	-0.003	(-1.70)	-0.003	(-1.64)
IV_{t-4}	0.000	(0.08)	0.000	(0.09)
$ r_t $	0.080	(7.94)	0.080	(7.93)
$ r_t \times I^{r_t>0}$	-0.049	(-4.72)	-0.049	(-4.72)
$ r_{t-1} $	0.065	(6.43)	0.064	(6.39)
$ r_{t-1} \times I^{r_{t-1}>0}$	-0.036	(-3.53)	-0.036	(-3.52)
$ r_{t-2} $	0.042	(5.00)	0.042	(4.89)
$ r_{t-2} \times I^{r_{t-2}>0}$	-0.035	(-3.89)	-0.035	(-3.87)
$ r_{t-3} $	0.041	(4.93)	0.041	(4.80)
$ r_{t-3} \times I^{r_{t-3}>0}$	-0.034	(-3.80)	-0.304	(-3.72)

Table 6**Regressions using a matched sample**

Stocks are sorted into terciles based on the absolute value of the first component of market maker position gamma, that is $|Hedge\Gamma_{t-\tau,t}|$, for each trading day. For each stock in the highest absolute gamma tercile, we select a control stock from among the stocks in the lowest absolute gamma tercile that has the closest return to the stock in the highest tercile from day -5 to day 0 . We then subtract the return $|r_{t+1}|$ of the control stock from the return $|r_{t+1}|$ of the stock in the highest absolute gamma tercile. The dependent variable $|r_{t+1}|_{diff}$ is the difference between the absolute of the stock in the highest absolute gamma tercile and its control stock.

	Market Maker 1990–2001		Market Maker 2002–2012	
Constant	–0.001	(–9.46)	0.002	(28.43)
$Hedge\Gamma_{t-\tau,t}$	–0.308	(–5.68)	–0.172	(–4.00)
$Net\Gamma_{t-\tau}$	–0.139	(–4.46)	–0.089	(–5.06)
$Info\Gamma_{t-\tau,t}$	–0.331	(–6.87)	–0.189	(–4.43)
$ r_t _{diff}$	0.145	(15.29)	0.273	(18.07)
$ r_{t-1} _{diff}$	0.059	(7.93)	0.077	(8.13)
$ r_{t-2} _{diff}$	0.045	(5.88)	0.055	(7.54)
$ r_{t-3} _{diff}$	0.039	(5.04)	0.044	(6.30)
$ r_{t-4} _{diff}$	0.042	(5.96)	0.043	(6.19)
$ r_{t-5} _{diff}$	0.037	(5.02)	0.055	(6.91)
$ r_{t-6} _{diff}$	0.035	(5.19)	0.054	(6.23)
$ r_{t-7} _{diff}$	0.041	(5.66)	0.052	(7.62)
$ r_{t-8} _{diff}$	0.044	(5.55)	0.060	(7.92)
$ r_{t-9} _{diff}$	0.049	(6.72)	0.056	(7.43)

Table 7**Regressions of absolute return on components of gamma assuming constant net open interest for each option series**

Results of estimating model (8) expressing $|r_{t+1}|$ in terms of the components of the net position gamma assuming net public open interest of each options series is one. The variable $Hedge\Gamma_{t-\tau,t_constantOI}$ is the change in option market maker position gamma due to stock price movement from day $t - \tau$ to t , $net\Gamma_{t-\tau_constantOI}$ is MM net gamma on day $t - \tau$, and $Info\Gamma_{t-\tau,t_constantOI}$ is the change in net gamma due to MM position changes from day $t - \tau$ to t . Standard errors are constructed from a covariance matrix for the average coefficients, which is formed by clustering observations by date and firm with common shocks, a method detailed in Thompson (2011). The associated t -statistics are reported in parentheses. The coefficient estimates on the variables related to lagged absolute returns from day $t - 9$ to $t - 5$ are omitted.

	1990–2001		2002–2012	
Constant	0.017	(30.28)	0.009	(27.35)
$Hedge\Gamma_{t-\tau,t_constantOI}$	19.270	(1.28)	12.300	(1.13)
$Net\Gamma_{t-\tau_constantOI}$	39.427	(1.36)	36.491	(1.77)
$Info\Gamma_{t-\tau,t_constantOI}$	1.378	(0.21)	1.216	(0.25)
$ r_t $	0.158	(17.23)	0.180	(18.87)
$ r_t \times I^{r_t > 0}$	-0.047	(-5.02)	-0.082	(-9.79)
$ r_{t-1} $	0.089	(11.65)	0.141	(15.47)
$ r_{t-1} \times I^{r_{t-1} > 0}$	-0.065	(-8.10)	-0.067	(-7.50)
$ r_{t-2} $	0.063	(8.70)	0.119	(12.14)
$ r_{t-2} \times I^{r_{t-2} > 0}$	-0.045	(-5.99)	-0.064	(-7.22)
$ r_{t-3} $	0.047	(6.76)	0.092	(10.80)
$ r_{t-3} \times I^{r_{t-3} > 0}$	-0.036	(-4.79)	-0.059	(-6.33)
$ r_{t-4} $	0.050	(7.49)	0.081	(9.17)
$ r_{t-4} \times I^{r_{t-4} > 0}$	-0.034	(-4.08)	-0.043	(-5.39)
...	...	...	...	...

Table 8**Regressions using components of unnormalized gamma**

Results of estimating model (8) expressing the absolute return $|r_{t+1}|$ in terms of the components of the unnormalized net position gamma of delta hedgers and lagged absolute returns. The first two columns of results report coefficients and t -statistics when market makers are assumed to the delta hedgers. The last two columns report the results when both market makers and firm proprietary traders are assumed to delta hedge. Standard errors are constructed from a covariance matrix for the average coefficients, which is formed by clustering observations by date and firm with common shocks, a method detailed in Thompson (2011). The associated t -statistics are reported in parentheses. The coefficient estimates on the variables related to lagged absolute returns from day $t - 9$ to $t - 5$ are omitted.

<i>A. 1990–2001</i>	Market Maker		Market Maker + Firm	
Constant	0.019	(32.47)	0.018	(32.51)
$unnorm[Hedge\Gamma_{t-\tau,t}] \times 10^{-6}$	-0.226	(-9.16)	-0.186	(-9.38)
$unnorm[Net\Gamma_{t-\tau}] \times 10^{-6}$	-0.210	(-8.15)	-0.183	(-6.62)
$unnorm[Info\Gamma_{t-\tau,t}] \times 10^{-6}$	-0.232	(-11.17)	-0.198	(-10.46)
$ r_t $	0.147	(17.28)	0.147	(17.39)
$ r_t \times I^{r_t > 0}$	-0.046	(-5.08)	-0.046	(-5.10)
$ r_{t-1} $	0.087	(11.82)	0.087	(11.90)
$ r_{t-1} \times I^{r_{t-1} > 0}$	-0.063	(-8.20)	-0.063	(-8.24)
$ r_{t-2} $	0.061	(8.76)	0.061	(8.82)
$ r_{t-2} \times I^{r_{t-2} > 0}$	-0.043	(-6.00)	-0.043	(-6.04)
$ r_{t-3} $	0.045	(6.77)	0.046	(6.82)
$ r_{t-3} \times I^{r_{t-3} > 0}$	-0.034	(-4.68)	-0.034	(-4.70)
$ r_{t-4} $	0.058	(8.52)	0.058	(8.58)
$ r_{t-4} \times I^{r_{t-4} > 0}$	-0.041	(-5.92)	-0.041	(-5.97)
...	...	...	...	...
<i>B. 2002–2012</i>				
Constant	0.009	(26.74)	0.009	(27.10)
$unnorm[Hedge\Gamma_{t-\tau,t}] \times 10^{-6}$	-0.163	(-6.72)	-0.123	(-5.54)
$unnorm[net\Gamma_t(t - \tau, S_{t-\tau})] \times 10^{-6}$	-0.120	(-5.45)	-0.083	(-8.43)
$unnorm[Info\Gamma_{t-\tau,t}] \times 10^{-6}$	-1.194	(-7.31)	-0.137	(-7.99)
$ r_t $	0.140	(13.50)	0.140	(13.72)
$ r_t \times I^{r_t > 0}$	-0.080	(-7.36)	-0.080	(-7.47)
$ r_{t-1} $	0.114	(14.72)	0.114	(14.94)
$ r_{t-1} \times I^{r_{t-1} > 0}$	-0.059	(-6.92)	-0.059	(-7.03)
$ r_{t-2} $	0.090	(12.49)	0.090	(12.70)
$ r_{t-2} \times I^{r_{t-2} > 0}$	-0.062	(-7.47)	-0.062	(-7.59)
$ r_{t-3} $	0.082	(11.44)	0.082	(11.61)
$ r_{t-3} \times I^{r_{t-3} > 0}$	-0.050	(-6.20)	-0.050	(-6.29)
$ r_{t-4} $	0.092	(11.56)	0.092	(11.73)
$ r_{t-4} \times I^{r_{t-4} > 0}$	-0.051	(-6.76)	-0.051	(-6.87)
...	...	...	...	...

Table 9

Regressions explaining the change in absolute return using components of gamma and lagged changes in absolute returns

The dependent variable is the first difference in the absolute return $\Delta |r_{t+1}| = |r_{t+1}| - |r_t|$. $Hedge\Gamma_{t-\tau,t}$ is the change in delta hedgers' net position gamma due to stock price movement from day $t - \tau$ to t , $Net\Gamma_t(t - \tau, S_{t-\tau})$ is the net position gamma on day $t - \tau$, and $Info\Gamma_{t-\tau,t}$ is the change in delta hedgers' net position gamma due to changes in open interest. The covariates $\Delta |r_{t-i}| = |r_{t-i}| - |r_{t-i-1}|$ for $i = 0, \dots, 9$ are the first differences in in lagged absolute returns, and $I^{r_t > 0}$ is an indicator variable for positive r_t . Standard errors are constructed from the covariance matrix of the average coefficients, which is formed by clustering observations by date and firm with common shocks. The associated t -statistics are reported in parentheses. The coefficient estimates on the variables related to lagged absolute returns from day $t - 9$ to $t - 5$ are omitted.

A. 1991–2001	Market Maker		Market Maker + Firm	
Constant	0.000	(1.77)	0.000	(1.68)
$Hedge\Gamma_{t-\tau,t}$	-0.369	(-4.26)	-0.317	(-3.69)
$Net\Gamma_{t-\tau}$	0.049	(2.44)	0.043	(2.19)
$Info\Gamma_{t-\tau,t}$	-0.398	(-6.85)	-0.345	(-6.07)
$\Delta r_t $	-0.788	(-46.76)	-0.788	(-46.78)
$\Delta r_t \times I^{r_t > 0}$	-0.037	(-2.98)	-0.038	(-2.99)
$\Delta r_{t-1} $	-0.671	(-35.49)	-0.670	(-35.49)
$\Delta r_{t-1} \times I^{r_{t-1} > 0}$	-0.043	(-3.51)	-0.044	(-3.52)
$\Delta r_{t-2} $	-0.580	(-29.75)	-0.579	(-29.77)
$\Delta r_{t-2} \times I^{r_{t-2} > 0}$	-0.030	(-2.82)	-0.030	(-3.83)
$\Delta r_{t-3} $	-0.495	(-25.60)	-0.495	(-25.62)
$\Delta r_{t-3} \times I^{r_{t-3} > 0}$	-0.026	(-2.52)	-0.026	(-2.53)
$\Delta r_{t-4} $	-0.401	(-22.37)	-0.400	(-22.37)
$\Delta r_{t-4} \times I^{r_{t-4} > 0}$	-0.030	(-2.77)	-0.030	(-2.78)
...	...	...	...	...
B. 2002–2012				
Constant	0.000	(1.64)	0.000	(1.80)
$Hedge\Gamma_{t-\tau,t}$	-0.214	(-3.68)	-0.181	(-4.27)
$Net\Gamma_{t-\tau}$	0.038	(3.42)	0.022	(2.98)
$Info\Gamma_{t-\tau,t}$	-0.188	(-7.53)	-0.185	(-6.97)
$\Delta r_t $	-0.829	(-61.40)	-0.829	(-62.17)
$\Delta r_t \times I^{r_t > 0}$	-0.048	(-3.87)	-0.048	(-3.93)
$\Delta r_{t-1} $	-0.711	(-52.14)	-0.711	(-52.78)
$\Delta r_{t-1} \times I^{r_{t-1} > 0}$	-0.028	(-2.61)	-0.028	(-2.65)
$\Delta r_{t-2} $	-0.608	(-43.05)	-0.607	(-43.58)
$\Delta r_{t-2} \times I^{r_{t-2} > 0}$	-0.036	(-3.33)	-0.036	(-3.38)
$\Delta r_{t-3} $	-0.519	(-38.59)	-0.519	(-39.11)
$\Delta r_{t-3} \times I^{r_{t-3} > 0}$	-0.024	(-2.39)	-0.024	(-2.43)
$\Delta r_{t-4} $	-0.417	(-33.68)	-0.416	(-34.14)
$\Delta r_{t-4} \times I^{r_{t-4} > 0}$	-0.025	(-2.38)	-0.025	(-2.41)
...	...	...	...	...

Table 10**Descriptive statistics for index returns and cross-sectional average gammas**

Means, standard deviations, extrema, and quantiles for the average gamma variables used to predict S&P 500 absolute returns. $AvgNet\Gamma_t$ is the average of the market makers' normalized net position gamma across the largest 500 stocks during the sample period. $AvgHedge\Gamma_{t-\tau,t}$ is the average of the component of $net\Gamma_t$ due to past stock price change, and $AvgInfo\Gamma_{t-\tau,t}$ the average of the component of $net\Gamma_t$ due to changes in options positions. The 1990–2001 and 2002–2012 samples include 3,023 and 2,764 observations, respectively.

	Mean	Std.	Min.	10%	25	50%	75%	90%	Max.
1990–2001									
$ r_t $	0.0071	0.0069	0.0000	0.0009	0.0022	0.0051	0.0096	0.0157	0.0687
$AvgNet\Gamma_{t-\tau,t}$	0.0036	0.0007	−0.0002	0.0019	0.0027	0.0036	0.0045	0.0054	0.0086
$AvgHedge\Gamma_{t-\tau,t}$	0.0000	0.0004	−0.0027	−0.0003	−0.0001	0.0000	0.0002	0.0004	0.0024
$AvgInfo\Gamma_{t-\tau,t}$	0.0000	0.0005	−0.0047	−0.0017	−0.0004	0.0003	0.0006	0.0009	0.0020
2002–2012									
$ r_t $	0.0089	0.0101	0.0000	0.0010	0.0024	0.0059	0.0117	0.0199	0.1158
$AvgNet\Gamma_{t-\tau,t}$	−0.0004	0.0016	−0.0041	−0.0011	−0.0006	−0.0002	0.0000	0.0001	0.0009
$AvgHedge\Gamma_{t-\tau,t}$	0.0000	0.0013	−0.0056	−0.0002	−0.0003	0.0000	0.0000	0.0002	0.0041
$AvgInfo\Gamma_{t-\tau,t}$	0.0000	0.0010	−0.0010	−0.0002	−0.0001	0.0000	0.0000	0.0001	0.0024

Table 11**Regressions of S&P 500 index absolute returns on cross-sectional averages of components of net position gamma**

Results of estimating model (9) expressing the absolute return of the S&P 500 index $|r_{t+1}^I|$ in terms of the cross-sectional averages of the components of the normalized net market maker position gammas and lagged returns. $I^{r_t^I > 0}$ is a dummy variable that equals one when $r_t^I > 0$. The model is estimated assuming that Market Makers delta hedge. Standard errors are constructed using the Newey-West method with ten lags. The associated t -statistics are in parentheses. The coefficient estimates on the variables related to lagged absolute returns from day $t - 9$ to $t - 5$ are omitted. The 1990–2001 and 2002–2012 samples include 3,023 and 2,764 observations, respectively.

	1990–2001		2002–2012	
Constant	0.004	(7.53)	0.002	(6.54)
$avgHedge\Gamma_{t-\tau,t}$	-0.808	(-2.69)	-0.476	(-2.11)
$avgNet\Gamma_{t-\tau}$	-0.331	(-3.29)	-0.262	(-1.40)
$avgInfo\Gamma_{t-\tau,t}$	-0.406	(-3.59)	-0.522	(-1.23)
$ r_t^I $	0.102	(2.62)	0.107	(2.01)
$ r_t^I \times I^{r_t^I > 0}$	-0.172	(-4.12)	-0.223	(-6.23)
$ r_{t-1}^I $	0.189	(5.81)	0.162	(3.41)
$ r_{t-1}^I \times I^{r_{t-1}^I > 0}$	-0.230	(-7.56)	-0.175	(-3.62)
$ r_{t-2}^I $	0.096	(3.52)	0.120	(3.14)
$ r_{t-2}^I \times I^{r_{t-2}^I > 0}$	-0.113	(-4.14)	-0.164	(-4.26)
$ r_{t-3}^I $	0.058	(2.11)	0.180	(4.18)
$ r_{t-3}^I \times I^{r_{t-3}^I > 0}$	-0.003	(-0.09)	-0.200	(-3.72)
$ r_{t-4}^I $	0.172	(5.70)	0.233	(5.92)
$ r_{t-4}^I \times I^{r_{t-4}^I > 0}$	-0.120	(-3.64)	-0.142	(-3.30)
...	...	...	...	...